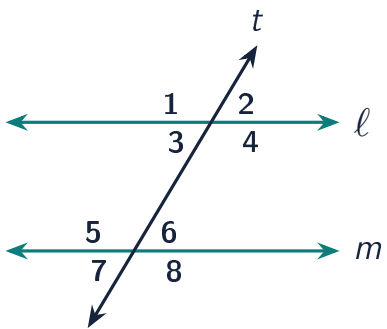
Lesson 3.2

# Parallel Lines & Transversals

Equal or supplementary — that's the whole pattern

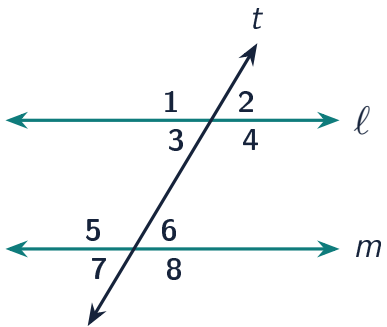
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# When $\ell \parallel m$




**Corresponding** angles are **congruent**.

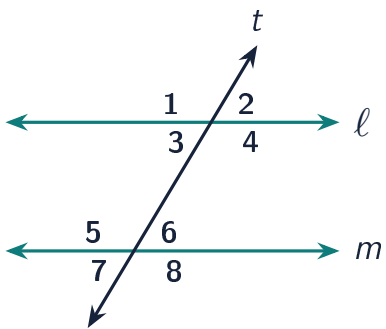
# When $\ell \parallel m$




**Corresponding** angles are **congruent**.

**Alternate interior** and **alternate exterior** angles are **congruent**.

# When $\ell \parallel m$

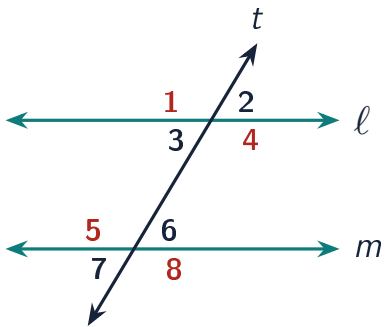



**Corresponding** angles are **congruent**.

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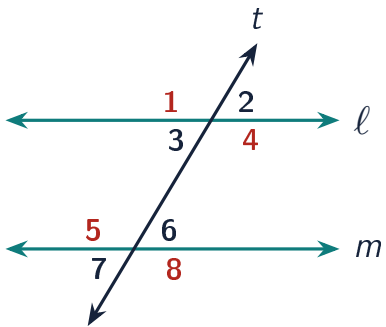
**Same-side interior** angles are **supplementary** (add to 180°).

# The easy way to see it




Every angle is either **acute** or **obtuse**.

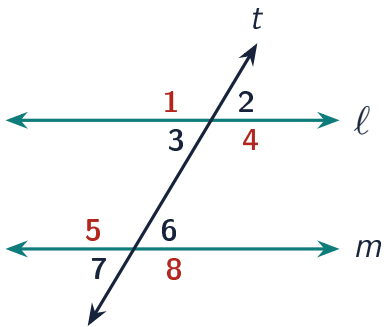
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Every angle is either **acute** or **obtuse**.

Two acute angles are **equal**. Two obtuse angles are **equal**.

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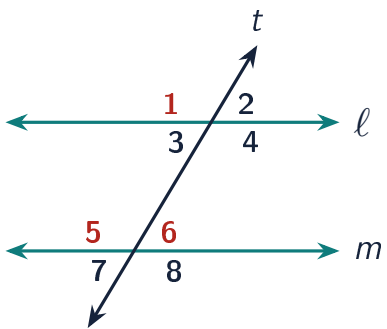



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Two acute angles are **equal**. Two obtuse angles are **equal**.

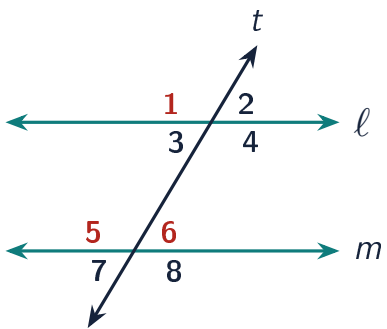
An acute + an obtuse = $180^\circ$

# Find the angle




$\ell \parallel m$, $m\angle 1 = 110°$. Find $m\angle 5$ and $m\angle 6$.

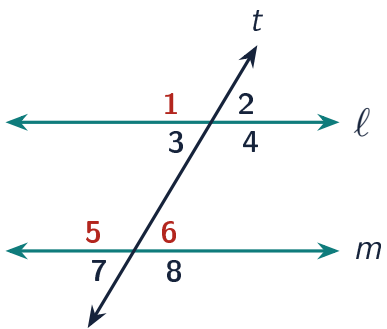
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$\angle 1$, $\angle 5$ corresponding:
$m\angle 5 = 110°$

$\angle 5$, $\angle 6$ linear pair:
$m\angle 6 = 180 - 110 = 70°$

# With algebra

Alternate interior: $(3x + 15)^\circ$ and $(5x - 25)^\circ$

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Congruent: $3x + 15 = 5x - 25 \Rightarrow 40 = 2x \Rightarrow x = 20$

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Alternate interior: $(3x + 15)^\circ$ and $(5x - 25)^\circ$

Congruent: $3x + 15 = 5x - 25 \Rightarrow 40 = 2x \Rightarrow x = 20$

Angle: $3(20) + 15 = 75^\circ$

# Watch out

**Watch out!** Same-side interior angles are **supplementary**, not equal. Set them to add to $180^\circ$.

**Your turn!** Pause the video and try it.

$\ell \parallel m$. Same-side interior angles are $(2x + 10)^\circ$ and $(3x - 5)^\circ$. Find $x$ and both angles.

**Your turn!** Pause the video and try it.

$\ell \parallel m$. Same-side interior angles are $(2x + 10)^\circ$ and $(3x - 5)^\circ$. Find $x$ and both angles.

$5x + 5 = 180 \Rightarrow x =$ 35; angles $80^\circ$ and $100^\circ$

# Remember

- Parallel lines: corresponding, alternate interior, alternate exterior angles are **congruent**.
- Same-side interior angles are **supplementary**.
- Acute $=$ acute, obtuse $=$ obtuse, acute $+$ obtuse $= 180°$.

**Now practice at your level:**

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